Ada in Wonderland

Input file:	`standard input`
Output file:	`standard output`
Time limit:	1 second
Memory limit:	256 megabytes

Ada decided to take a vacation and go on an unforgettable trip to Wonderland! Upon arriving, she realized that she would need to convert all her money to the local currency to be able to enjoy her stay.

She went to an exchange office and found out that 1 coin from her home country is equivalent to exactly Y coins in Wonderland's monetary system.



Knowing that Ada packed X coins from her country, your task is to help her calculate how many coins she will have to spend in Wonderland after making the exchange.

Input

The input consists of two lines, each containing an integer.

The first line contains the integer X ($1 \leq X \leq 1000$), which represents the amount of coins Ada brought on her trip.

The second line contains the integer Y ($1 \leq Y \leq 1000$), which is the value corresponding to the currency exchange rate (how many Wonderland coins are equivalent to 1 coin from Ada's country).

Output

The output should contain a single integer, which represents the total value, in coins, that Ada possesses according to Wonderland's monetary system.

Examples

standard input	standard output
1 1	1
3 5	15
10 20	200

Lexicographical Battle

Input file:	`standard input`
Output file:	`standard output`
Time limit:	3 seconds
Memory limit:	256 megabytes

Ada and Py meet again after a long time and decide to play a game, which will be mediated by a judge.

At the beginning of the game, Ada has a non-empty string s_A, while Py has a string s_P, which is also non-empty. The game consists of a sequence of Q operations, which must be processed in order. These operations can be of 3 different types (1, 2 or 3), described below:

1. Given an integer k and a string s, append k copies of the string s to the end of the string s_A. In other words, after an operation of type (1 k s), the string s_A becomes equal to the concatenation of s_A with k copies of the string s.
2. Given an integer k and a string s, append k copies of the string s to the end of the string s_P. In other words, after an operation of type (2 k s), the string s_P becomes equal to the concatenation of s_P with k copies of the string s.
3. Indicates a round of the game. In each round, Ada and Py are asked to send their strings (s_A and s_P, respectively) to the judge. Before sending them, both Ada and Py can reorder their strings however they want. In other words, Ada can send any permutation of s_A, while Py can send any permutation of the string s_P. Upon receiving the two strings, the judge declares the winner of the round as the one who sent the lexicographically smaller string. If both strings are equal, the judge declares a tie as the result of the round.

It is guaranteed that s_A and s_P initially consist only of lowercase English letters, as do all strings s present in operations of types 1 and 2.

We say that a string A is lexicographically smaller than a string B if, and only if, one of the following two conditions is satisfied:

1. A is a prefix of B and A is different from B.
2. At the first position where A and B differ, A has a letter that precedes the corresponding letter in B in alphabetical order.



Your task is, for each operation of type 3, to output the result determined by the judge considering that Ada and Py play optimally (considering the strings s_A and s_P at the moment the round occurs).

Input

The first line of the input contains the non-empty string s_A at the beginning of the game.

The second line contains the non-empty string s_P at the beginning of the game.

The third line contains an integer Q ($1 \leq Q \leq 10^5$), indicating the number of operations that will occur throughout the game.

The next Q lines start with an integer t ($1 \leq t \leq 3$), indicating the type of that operation. If $t = 1$ or $t = 2$, t is followed by an integer k ($1 \leq k \leq 10^5$) and a non-empty string s.

It is guaranteed that all strings in the input contain only lowercase English letters and that the sum of the lengths of all of them does not exceed $2 \cdot 10^5$.

Output

For each operation of type 3, print `ADA` if Ada is the winner of the round, `PY` if Py is the winner of the round, or `EMPATE` if the judge determines that the round in question ended in a tie.

Examples

standard input	standard output
mfp mfp 7 3 2 1 s 3 2 1 f 3 1 2 fps 3	EMPATE ADA PY ADA
ada py 5 1 100000 ada 1 100000 py 1 100000 ada 2 1 aba 3	ADA

Cutting Bookmarks

Input file:	`standard input`
Output file:	`standard output`
Time limit:	1 second
Memory limit:	256 megabytes

Amy owns a small stationery shop in the city center. Recently, she developed a very special new bookmark model that is a huge success among students.

These bookmarks are made of a resistant and colorful material, cut from a large rectangular sheet that Amy buys from her supplier. Each rectangular sheet is exactly X centimeters wide by Y centimeters long.

To optimize her production and avoid waste, Amy cuts the sheet into bookmarks that are exactly 1 centimeter wide by 7 centimeters long. Due to technical constraints of the cutting machine Amy uses, she must choose only one orientation for all bookmarks on each sheet — either all aligned horizontally or all aligned vertically with the edges of the sheet.

At the end of the day, Amy needs to know how many complete bookmarks she can produce from each rectangular sheet, considering the best possible orientation. She does not consider incomplete pieces, as they cannot be sold.



Since Amy is very busy attending to customers, she asked for your help to create a program that quickly calculates the maximum number of bookmarks that can be produced from a rectangular sheet with dimensions X and Y, choosing the orientation that maximizes production.

Input

The input consists of a single line containing two integers X and Y ($1 \leq$ X, Y ≤ 1000), representing, respectively, the width and the length of the rectangular sheet in centimeters.

Output

Print a single integer representing the maximum number of complete bookmarks (1 cm $\times$ 7 cm) that can be produced.

Examples

standard input	standard output
1 10	1
700 100	10000

Dividing Stickers

Input file:	`standard input`
Output file:	`standard output`
Time limit:	1 second
Memory limit:	256 megabytes

Ada is a big football fan and is very excited about completing her World Cup sticker album. After buying and opening hundreds of packets, she finally managed to complete the album, but ended up accumulating several duplicate stickers.

To organize her collection, Ada separated her duplicate stickers by country. Since there are N different national teams participating in the World Cup, she made N stacks, so that each stack contains only stickers from the same team. The i-th stack has exactly A_i stickers.

Not seeing any sense in hoarding duplicate stickers, Ada decided to distribute them among her friends. In order to maintain organization and fairness in the division, the following conditions must be met:

- Each chosen friend must receive stickers from a single team (i.e., stickers that are in the same stack).
- All chosen friends must receive the same amount of stickers.
- All of Ada's stickers must be distributed (i.e., no stickers should remain with Ada).



Ada wants each friend to receive the maximum possible amount of stickers. Consequently, this means she needs to choose the minimum possible number of friends to receive the donation, respecting all the rules above.

Your task is to help Ada calculate the minimum number of friends who can be gifted with her duplicate stickers.

Input

The input consists of two lines.

The first line contains an integer N ($1 \leq N \leq 10^5$), representing the number of countries participating in the World Cup (and, consequently, the number of Ada's stacks of duplicate stickers).

The second line contains N space-separated integers $A_1, A_2, \ldots, A_N$ ($1 \leq A_i \leq 10^9$), where A_i represents the amount of duplicate stickers for the i-th country.

Output

The output must contain a single integer: the minimum number of friends who can be chosen to receive Ada's stickers, obeying the described conditions.

Examples

standard input	standard output
1 10	1
3 10 15 7	32
4 24 60 36 48	14

Note

In the first test case, Ada has duplicate stickers from a single country, totaling 10 stickers. She can give all 10 stickers to a single friend according to the established rules.

In the second test case, Ada has piles with 10, 15, and 7 duplicate stickers. The only division that respects the imposed conditions consists of giving 1 sticker to each friend. Therefore, she will need to gift $10 + 15 + 7 = 32$ friends.

Tire Strategy

Input file:	`standard input`
Output file:	`standard output`
Time limit:	1 second
Memory limit:	256 megabytes

The 2026 season of MFP Racing, one of the most important motorsport competitions in the world, is about to begin. You have been hired as a consultant for one of the teams and your first task is to develop a program that creates a tire strategy for the opening race, minimizing the total race time.

There are three types of tires available for the drivers:

- Soft — fast, but lasts fewer laps.
- Medium — balanced.
- Hard — slower, but has higher durability.

Each type of tire has a specific time spent per lap and a maximum durability. In addition, each race has N laps. The car must complete all of them, and it is allowed to start with any type of tire.

During the race, tire changes (pit-stops) can be made in the pits. These stops always occur after the completion of a lap and are not counted as race laps (that is, making a pit-stop does not reduce the number of laps that still need to be driven). Thanks to the great efficiency of the team's mechanics, you already know that each stop adds a fixed time P, in seconds, to the total race time. However, there is a limit of at most S pit-stops per race.



Your task is to calculate the minimum time needed to finish the race, respecting the exchange limit and the characteristics of each tire. It is guaranteed that there is always at least one strategy with which it is possible to finish the race.

Input

The input consists of three lines.

The first line contains three integers N, P and S ($2 \leq N \leq 1000, 1 \leq P \leq 100, 0 \leq S \leq 5$), representing, respectively, the number of laps, the time (in seconds) spent on each tire change, and the maximum number of allowed changes.

The second line contains three integers t_1, t_2 and t_3 ($1 \leq t_1 < t_2 < t_3 \leq 300$), representing the time spent per lap (in seconds) for each type of tire (soft, medium and hard, respectively).

The third line contains three integers d_1, d_2 and d_3 ($1 \leq d_1 < d_2 < d_3 \leq N$), representing the maximum number of laps that each type of tire (soft, medium and hard, respectively) can endure before a pit-stop is required.

Output

Print a single integer: the **minimum possible total time**, in seconds, to complete the race.

Example

standard input	standard output
10 20 2 80 83 90 4 6 7	838

Note

The best strategy for the example case consists of:

- 4 laps with the soft tire ($4 \times 80 = 320$ s)
- pit-stop (20 s)
- 6 laps with the medium tire ($6 \times 83 = 498$ s)
- total $= 320 + 20 + 498 =$ **838 seconds**.

Dungeon Escape

Input file:	`standard input`
Output file:	`standard output`
Time limit:	1 second
Memory limit:	256 megabytes

Ada is a bird who loves gathering with her forest friends to play RPG, a game where they create stories together and use dice to decide the success of their actions. In today's session, Ada's character finds herself in a tough situation: while exploring an ancient underground dungeon, she triggered a trap, and now the ceiling is rapidly descending upon her group.

To escape this situation, the Game Master, the friend who organizes and dictates the rules of the game, warns that Ada needs to pass a luck test. The test consists of rolling N different dice simultaneously. The i-th die has d_i faces, so, when rolling it, Ada can obtain, with equal probability, any integer in the range $[1, d_i]$.

The rule imposed by the Game Master is simple: the group will only manage to save themselves if the highest value obtained, considering all N rolled dice, is greater than or equal to a value K.



Ada would like to know her real chances of survival before rolling the dice. Given the number of dice, the number of faces on each one, and the value K stipulated by the Game Master, your task is to calculate the probability of Ada and her group successfully escaping the trap.

Input

The input consists of two lines.

The first line contains two integers N, K ($1 \leq N \leq K \leq 100$), separated by a space, representing, respectively, the number of dice Ada must roll and the value established by the Game Master for the test.

The second line contains N space-separated integers d_i ($2 \leq d_i \leq 100$), representing the number of faces of each of the N dice.

It is guaranteed that $K \leq d_i$ for some $1 \leq i \leq N$.

Output

The output must contain a single real number r such that $0 \leq r \leq 1$, representing the probability of Ada passing the test.

Your answer will be considered correct if the relative or absolute difference from the judge's answer is up to 10^{-6}.

Hint: To print a floating-point number r with precision in Python, you can use:

`print(f"{r:.10f}")`

In C++, you can use `setprecision`:

```
std::cout << std::fixed << std::setprecision(10) << r << "\n";
```

Examples

standard input	standard output
1 2 2	0.500000000000000
1 5 5	0.200000000000000
2 3 3 3	0.555555555555556
3 4 5 2 3	0.400000000000000

Win or Lose?

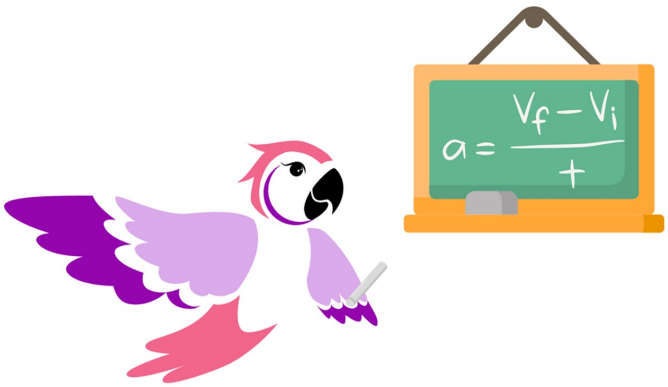
Input file:	`standard input`
Output file:	`standard output`
Time limit:	1 second
Memory limit:	256 megabytes

Ada is a student fascinated by strategy games and was challenged by her Physics professor to a match of a numerical game.

The game begins with a positive integer N written on the board. The two players alternate turns and, being the challenged one, Ada has the right to make the first move. In each turn, the current player must choose a number that is a power of 2 (i.e., 1, 2, 4, 8, 16, ...) and subtract this value from the number written on the board, erasing the old number and writing the new result. It is only allowed to subtract a value that is less than or equal to the current number on the board.

Formally, if the number written on the board is x, the current player can choose an integer $k \geq 0$ such that $2^k \leq x$ and replace x with $x - 2^k$.

The player who makes the move that reduces the number on the board to exactly 0 wins.



Ada is extremely competitive and hates to lose. Before accepting the professor's challenge, she wants to analyze the initial number N and find out if she has a guaranteed victory or if defeat is inevitable (assuming that both she and the professor will play optimally).

Write a program that, given the number N written on the board, answers the question stuck in Ada's head: "Win or Lose?"

Input

The only line of the input contains a single integer N ($1 \leq N \leq 10^{18}$), indicating the number written on the board at the beginning of the game.

Output

Print `GANHA` if Ada has a winning strategy playing first. Otherwise (i.e., if the professor has a winning strategy playing second), print `PERDE`.

Examples

standard input	standard output
2	GANHA
30	PERDE
1000000000000000	GANHA

Time to Walk

Input file:	`standard input`
Output file:	`standard output`
Time limit:	2 seconds
Memory limit:	512 megabytes

The final phase of the Latin American Women's Programming Contest (MFP) will take place once again at Unicamp in 2026. To welcome the finalists in the best possible way, the organization has prepared a map highlighting N crucial points of interest on campus (such as the Institute of Computing and the University Restaurant).

To ensure no participant gets lost, the organization mapped exactly $N - 1$ main paths connecting these locations, forming a structure in which there is a single route between any pair of points. Each path has an estimated walking time in minutes.

However, the Unicamp campus is a very dynamic environment. The walking time of a path can change suddenly during the day due to unforeseen events such as puddles after quick rain showers, huge queues that form and turn the corner near the restaurants, or even a family of capybaras that decided to take a nap in the middle of the sidewalk!

To guarantee the well-being of the finalists, the MFP coordination wants to monitor the "Fatigue Factor" of the campus, which is defined as the sum of the walking times, in minutes, between all possible pairs of points of interest. Every time an unforeseen event occurs and the walking time of a path is altered, the coordination needs this factor to be immediately recalculated.



Given the map prepared by the MFP organization, your task is to process Q updates to path walking times throughout the day. After each change, you must calculate and print the updated Fatigue Factor. Since the participants will walk a lot and the sum of the times can be very large, print the results modulo $10^9 + 7$.

Input

The first input line contains two integers N, Q ($2 \leq N \leq 2 \cdot 10^5$, $1 \leq Q \leq 2 \cdot 10^5$), representing, respectively, the number of points of interest on campus and the number of updates to path walking times throughout the day. The points of interest are numbered as $1, 2, \ldots, N$.

Next, there are $N - 1$ lines describing the paths mapped by the organization. The i-th of these lines contains three integers u, v and w ($1 \leq u, v \leq N$ and $1 \leq w \leq 10^9$), indicating that there is a path between points u and v with an initial walking time equal to w minutes. It is guaranteed that the paths form a structure in which there is exactly one route between any pair of points (a tree).

Finally, there are Q lines describing the updates resulting from the unforeseen events. Each line contains two integers j and x ($1 \leq j \leq N - 1$ and $1 \leq x \leq 10^9$), indicating that the walking time of the j-th path (in the order they were given in the input, starting from index 1) is changed to x minutes.

Output

Print Q integers: the updated "Fatigue Factor" (the sum of the walking times between all pairs of points of interest), calculated modulo $10^9 + 7$ after processing each update, printing one per line.

A value modulo $10^9 + 7$ is equal to the remainder of the division of the original value by $10^9 + 7$.

Example

standard input	standard output
5 4 1 2 1 1 3 2 2 4 3 2 5 4 1 6 2 19 4 1 2 1	72 140 128 56

Note

The image on the left represents the initial path configuration. The image on the right illustrates the campus structure after the first unforeseen event, which changes the walking time between points 1 and 2 from 1 to 6 minutes.

